

WOLGANE TCHAPTCHET

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Highly skilled **Electronics & Embedded Systems Engineer** specializing in **critical embedded software, Integration & Verification (I&V), low-level development, and avionics protocols**. Proven hands-on experience with the **V-Model (SDLC)**, hardware-in-the-loop (HIL) testing, troubleshooting, and multi-core architectures. Having worked in **DO-178C (Design Assurance Level DAL A-E)** environments on flight control computers, I am looking to join a leading global player to contribute to real-time software integration while ensuring safety, performance, and compliance for next-generation programs.

PROFESSIONAL EXPERIENCES

Embedded Software & Avionics Engineer Intern, Safran Electronics & Défense | Massy, France

March 2026 – August 2026

- **Critical Software Loading:** Developed and integrated a critical software-loading tool on Windows 11 for the e-VTOL flight control computer's Embedded Dataloader, complying with the **DO-178C** standard.
- **Hybrid Architecture Design:** Created an asynchronous proxy (Python/C++) bridging encapsulation and transport protocols (ARINC 615A/665, TFTP, UDP, IP, PPP) to the physical **RS-422 serial link** connected to the avionics computer.
- **Low-Level Dev & Optimization:** Designed the "**PPP-Blinky**" C++ library (native PPP implementation replacing the heavy Linux daemon) and performed memory refactoring, reducing data transfer times by **70%**.
- **Validation & Deployment:** Designed the Python-based graphical user interface (GUI), executed hardware-in-the-loop testing, compiled simulation builds (g++), conducted hardware/network diagnostics (**Wireshark, oscilloscope**), and wrote technical documentation in English.

Web & IoT Development Engineer Intern, AJVI-AJVC | Paris, France

June 2024 – September 2024

- Developed an IoT monitoring and supervision interface utilizing **SQLite** and **MySQL**.
- Secured data transmission channels using **HTTPS, SSL**, and **MQTT** protocols.
- Optimized system performance and successfully integrated embedded data streams.

Electronics Maintenance Intern, NT FOODS | Yaounde, Cameroon

June 2022 – August 2022

- Diagnosed, troubleshooted, and maintained industrial electronic control boards.
- Automated and optimized production line processes to increase manufacturing efficiency.

EDUCATION

ESEO Graduate School of Engineering, Angers, France

August 2023 – August 2026

- **Master of Science (MSc) in Engineering** – Electronics & Connected Objects (IoT)
- **Key Coursework:** Embedded Systems, RF, Low-Level Security, IoT, Android Native Development, PCB Design, Microwaves, CMOS, Electromagnetic Compatibility (EMC).

ENGINEERING PROJECTS

Smart Air Control – Connected Ventilation (Capstone Project) – 2026, ESEO | Angers, France

- **Software Lead** (équipe de 4) : Guided project delivery and developed a native Android application (Kotlin/Compose) for Indoor Air Quality (IAQ) monitoring, remote actuator control, and BLE-based alerting (**nRF52**).
- **Low-Level Development:** Wrote drivers for environmental sensors (**BME680**), controlled linear actuators, automated system testing, validated data packets using Docklight, and executed complete hardware/software integration.

IoT Connected Clock (BI-ESP Project) – 2025, ESEO | Angers, France

- **C++ Firmware:** Developed real-time firmware, managed communication protocols, integrated multiple sensors (**BME680, GPS NEO-6M**), and implemented NTP time synchronization.
- **Hardware Design:** Designed the complete PCB on Altium Designer, created 3D CAD models, and executed robustness testing on physical boards.

SKILLS & LANGUAGES / INTERESTS

Languages: **French:** Native, **English** (Professional Working Proficiency), **Spanish:** Beginner

Interests: Football (Soccer), automotive engineering, music, traveling, and photography.

Soft Skills: Team collaboration, Requirements analysis, Organization, Technical communication, Project leadership.

Embedded Software & Critical Systems: Embedded C / C++, Python, Integration & Verification (I&V), V-Model (SDLC), RTOS, Embedded Linux; **Avionics & Standards:** ARINC 615A, ARINC 665, DO-178C, Design Assurance Level (DAL A-E), Secure Communications; **Testing & Diagnostics:** Hardware-in-the-loop (HIL) testing, troubleshooting, bus analyzers, oscilloscopes, emulators, configuration management, DevOps (CI/CD).

Electronics & Hardware: STM32, ESP, nRF52, Sensor Drivers (BME680, GPS NEO-6M, Ultrasonic); **PCB & RF/IoT Design:** Altium, Cadence, EMC, BLE, Wi-Fi, RFID, RS-422, System Validation.

Volunteering: “Virades de l'Espoir” 2023, 2024 & 2025 | Lac de Maine – Angers, France

Supported event logistics, managed team coordination, and raised environmental awareness (waste collection) during public fundraising events.